Amith Ramdas Achari

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EDUCATION

University of Illinois at Urbana-Champaign | *Master's Mechanical Engineering - Robotics | CGPA: 3.81/4.00* *Dec 2022*
RV College of Engineering - India | *Bachelor's Mechanical Engineering | CGPA: 8.90/10.0* *July 2020*

WORK EXPERIENCE

Machine Simulation Engineer **- Caterpillar – Physical AI: Autonomy** *Jan 2023 - Present | Champaign, Illinois*

- *Ranked Top 2% of the engineering workforce with a '**Superior**' rating, **recognized** personally by the **Senior VP***
- *Selected for an International Technical Assignment to the India Tech Center to train engineers and standardize workflows*
- *Won the Spring 2025 Innovation Challenge among 400+ submissions; finalist in Fall 2025 and Spring 2026 challenges*

✦ ***Physical AI – GPU-Accelerated Machine Simulation (NVIDIA Newton / Isaac Sim)***

- Led a 4-engineer team with no prior Newton experience to deliver the org's first AT745 truck model in NVIDIA Newton
- Built GPU-parallel engine, transmission, steering, hoist and brake models in Python/Warp, validated against Dynasty
- Ported Simulink controls to GPU via AI-assisted Simulink → C → Warp kernel conversion, avoiding manual rewrites
- Scaled to 500 trucks simulated concurrently at 2.03x real time; benchmarked PhysX, MuJoCo, XPBD and Kamino solvers
- Architected an agentic AI pipeline generating validated sim-ready models from Dynasty/CAD, cutting months to days
- Delivered a closed-loop autonomy demo integrating terrain, tire and sensor models with the autonomy stack across 4 teams
- Automated Robotics dataset generation by replaying Dynasty results in Isaac Sim and exporting ROS bags

✦ ***Autonomy Middleware Engineer - Rotation***

- Engineered high-fidelity C++ virtual sensor framework emulating Honeywell HGuide IMU3 and BCam2 hardware sensors
- Integrated virtual sensors into the autonomy stack and deployed them on A7Z1 vECUs and Linux targets
- Implemented real-time sensor streaming (<5 ms) and command responses using socket programming and GStreamer pipelines
- Designed fault-injection architecture (saturation, corruption, dropouts), uncovering 15+ critical driver defects

✦ ***Accelerated Autonomy S/W Validation in SIL Infrastructure***

- Championed "left-shift" SIL validation with vECUs replacing physical HIL dependencies, increasing velocity by 35%
- Integrated vECU DLLs, plant models, sensor sim and autonomy stack end to end, cutting test-bench reliance by ~30%
- Developed custom CAN drivers (Vector/SocketCAN) for CAN-over-Ethernet between Windows vECUs and Linux autonomy

✦ ***Real-Time Plant Modeling and Hybrid HIL/SIL Validation***

- Architected Hybrid HIL/SIL platform coupling plant models with the autonomy stack across 500+ regression scenarios
- Converted high-fidelity models to real-time HIL/SIL plant models for 20+ NPI programs, running up to 10x faster
- Developed a reusable EH steering real-time module configurable to 50 machine variants, adopted on 7+ programs
- Root-caused 50+ issues across HIL benches (US/India), incl. a systemic engine defect spanning 5 machine models
- Built the first Dynasty–vSET real-time co-simulation; automation scripts saved ~10 hrs per software release
- Built ERS HIL co-simulation with multi-rate motor control DLLs and PLC links over OPC-UA and TCP/IP (Snap7)
- Grew the team by 3 engineers via recruiting and mentoring, ramping them up to own model deliveries independently

Powertrain Engineer **- Team Helios Racing, official BAJA Club** *Sept 2016 - May 2020 | Bengaluru, India*

- **Led** the team to become the **National Champions** at Enduro Student India, among 60 teams from all over the nation
- Designed and validated a 'force-multiplication' CVT mechanism, verifying MATLAB simulations against rig test data

PROJECT HIGHLIGHTS

Path Planning for Autonomous Navigation [GitHub] | *Feb 2022-May 2022*

- Implemented Dubins-constrained RRT* and Hybrid A* planning algorithms for autonomous GEM vehicle navigation
- Integrated planner with Model Predictive Controller (MPC) for trajectory tracking while considering non-holonomic constraints

Autonomous Pipeline [GitHub] | *Jan 2022-May 2022*

- Integrated perception, decision-making and PD control modules for lane tracking and obstacle avoidance on a racetrack
- Architected a modular C++ decision-making node fusing LiDAR and camera data for dynamic obstacle avoidance

Impedance & Position control on a Catalyst 5 robot arm [GitHub] | *Jan 2022-May 2022*

- Implemented task-space PD control with friction compensation to avoid obstacles and follow the optimized trajectory
- Demonstrated force-controlled manipulation (~50N) on fragile objects without causing structural damage

SKILLS

Languages/Tools:	C++, Python, MATLAB, Bash, Linux, Git, CMake, pytest, Azure DevOps
Software/Libraries:	NVIDIA Isaac Sim, Newton, Warp, MuJoCo, ROS, AIS, Carla, vSET, GStreamer, UDP/TCP
Modeling and Controls:	Simulink, Dynasty, dSPACE ControlDesk, CANape, CANalyzer, Vector CAN, SocketCAN